

RTAB-Map Localization and Navigation

RTAB-Map Localization and Navigation

1. Course Content
2. Principle Introduction
3. Start the Agent
4. Run the Example
 - 4.1 Preparations
 - 4.2 Start the Navigation Node
5. Source Code Analysis
 - 5.1 View TF Tree
 - 5.2 View Node Communication Graph
 - 5.3 Launch File

1. Course Content

1. Run the sample program to perform navigation using RTAB-Map in localization mode.

2. Principle Introduction

- The default localization algorithm in Navigation 2 is AMCL (Adaptive Monte Carlo Localization). This involves replacing the default AMCL localization method with the RTAB-Map localization method, while other navigation functions remain unchanged.

3. Start the Agent

Note: If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```
mircoROS_Agent
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1765183764.822476] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1765183764.826313] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x10D9EFBC, topic_id: 0x006(2), participant_
id: 0x000(1)
[1765183764.829482] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1765183764.834154] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1765183764.838233] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x10D9EFBC, topic_id: 0x007(2), participant_
id: 0x000(1)
[1765183764.842504] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1765183764.847468] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

4. Run the Example

4.1 Preparations

- You need to have a map in `.db` format built first. Refer to the [15-RTAB-Map SLAM Mapping](#) section.

4.2 Start the Navigation Node

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

Open a terminal inside the container,

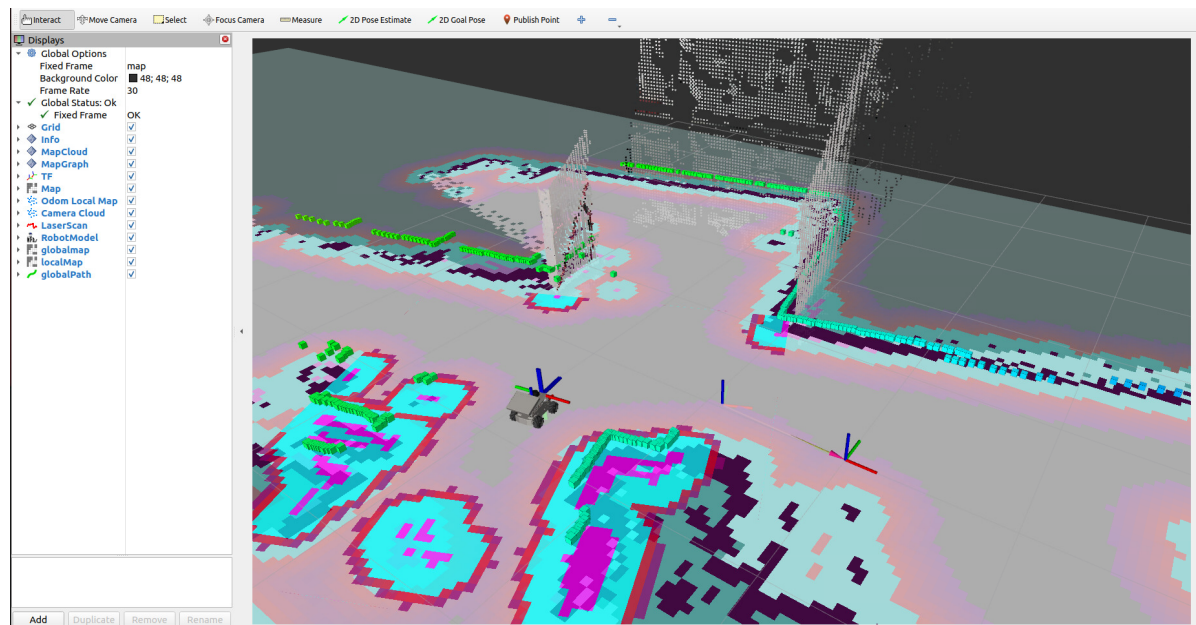
```
exec_m3
```

(ORIN board)

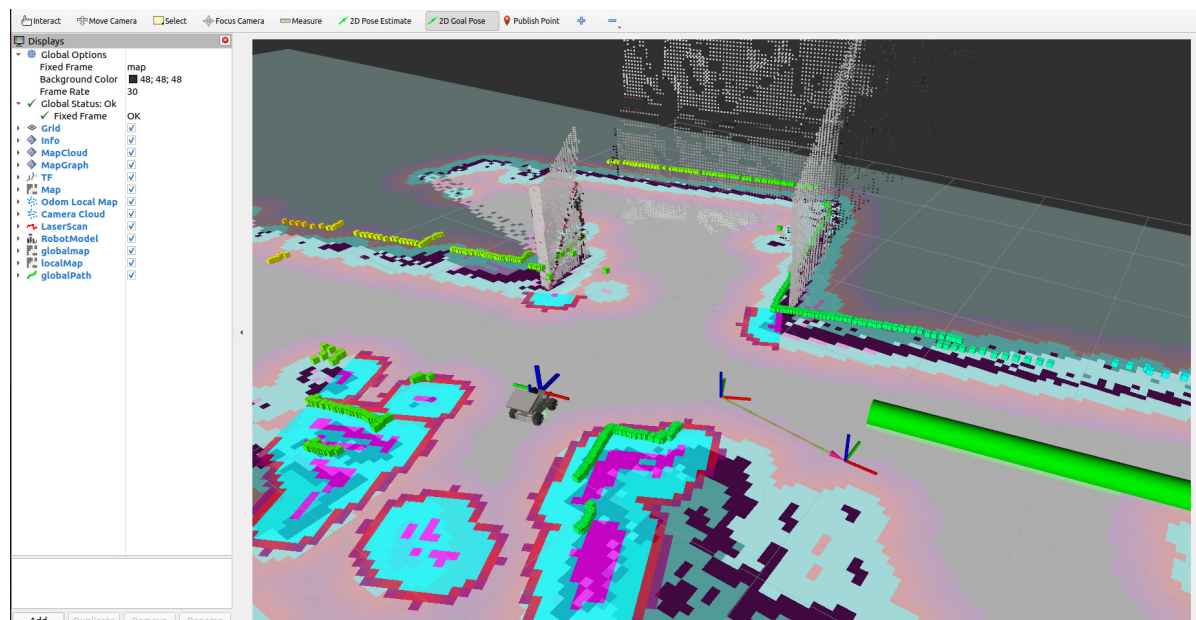
```
ros2 launch m3_bringup nav2.launch.py localization_mode:=rtabmap
```

(RDK & PI5 boards) Run in separate steps, with visualization launched on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`,

```
# On the virtual machine
rviz2 -d /home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/rtabmap_nav.rviz
# On the host machine
ros2 launch m3_bringup nav2.launch.py localization_mode:=rtabmap use_rviz:=False
```



- Navigation is also performed by giving a target pose with `2D Goal Pose`.

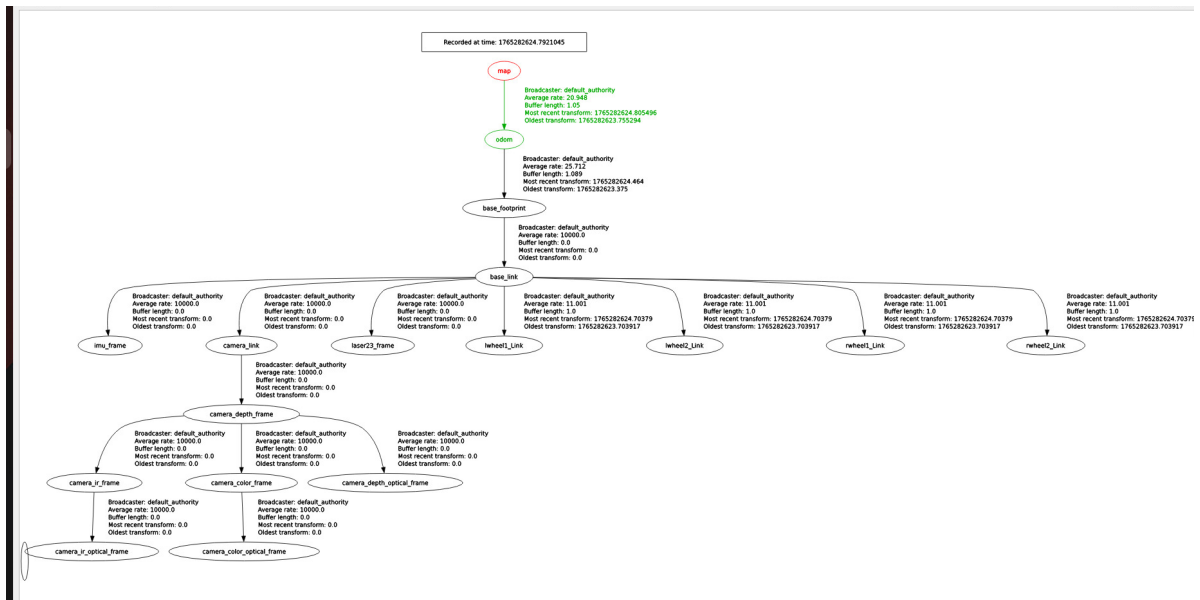


5. Source Code Analysis

5.1 View TF Tree

```
ros2 run rqt_tf_tree rqt_tf_tree
```

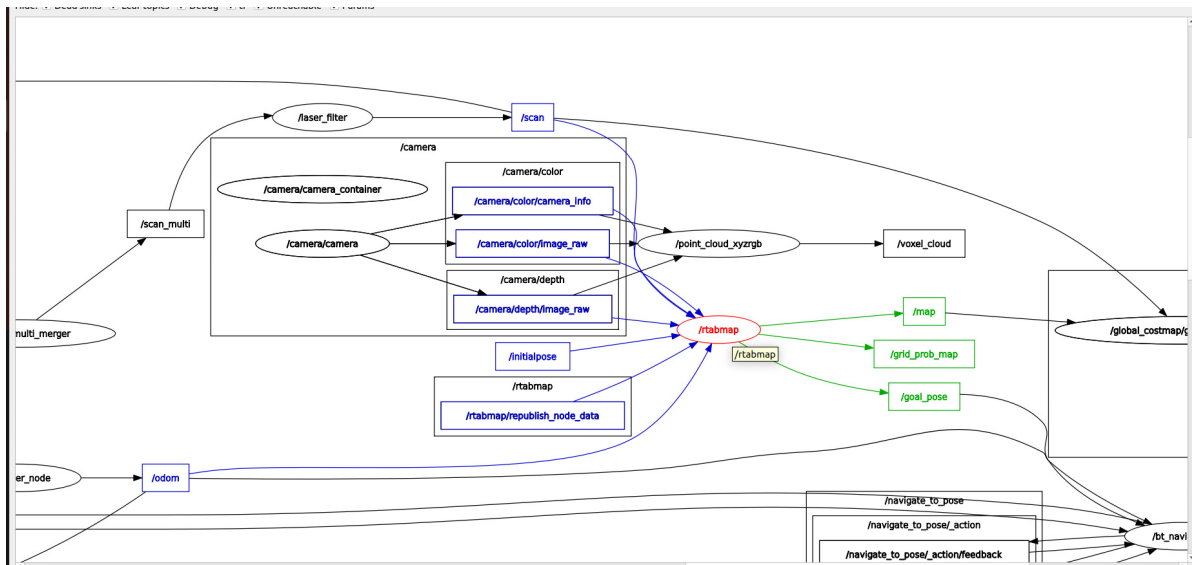
- Here, the TF transformation from the `map` coordinate frame to the `odom` coordinate frame is provided by the `rtabmap` node.



5.2 View Node Communication Graph

```
ros2 run rqt_graph rqt_graph
```

- In the figure below, when the `/rtabmap` node is in localization mode, it receives the Lidar `/scan` and odometry `/odom` topics, and publishes TF transformations and updates the `/map`.



5.3 Launch File

Source code path:

```
~/yahboomM3_ws/user_ws/src/m3_bringup/launch/navigation/nav2.launch.py
```

- The key code snippet below starts the Navigation2 stack node and the RTAB-Map localization node.

```
# RTAB-Map positioning navigation
rtabmap_localization_nav = GroupAction(
    condition=IfCondition(PythonExpression(["'", localization_mode, "' ==
'rtabmap' and '", slam, "' == 'False' "]])),
    actions=[
        PushRosNamespace(namespace=namespace),
        # Start navigation2 navigation file (no global positioning)
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource(nav2_launch_mod),
            launch_arguments={
                "map": map_dir,
                "params_file": params_file,
                "use_sim_time": use_sim_time,
            }.items()),
        # Start slamtoolbox positioning node
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource(rtabmap_localization_launch_file),
            launch_arguments={
                "mode": "localization",
            }.items(),
        ),
        # Log Printing
        LogInfo(msg=[Fore.GREEN+'=====Robot-M3 Navigation
Start with rtabmap localization====='], Fore.RESET])
    ]
)
```

- RTAB-Map localization node launch file path:

```
~/yahboomM3_ws/user_ws/src/m3_bringup/launch/slam/rtab_map.launch.py
```

- The key launch snippet below switches to localization mode via the `localization` launch parameter.

```
delay_start_rtabmap_localization= TimerAction(
    condition=IfCondition(PythonExpression(["'", mode, "' == 'localization'
])),
    period=5.0,
    actions=[
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource(rtabmap_launch),
```

```

launch_arguments={
    "rgb_topic":                rgb_topic,
    "depth_topic":              depth_topic,
    "camera_info_topic":        camera_info_topic,
    "frame_id":                  'base_link',
    "odom_topic":                '/odom',
    "use_sim_time":              'false',
    "rviz":                      'true',
    "rviz_cfg":                  nav_cfg,
    "database_path":             database_path,
    "rtabmap_viz":               'false',
    "approx_sync":               'true',
    "approx_sync_max_interval": '0.01',
    "qos":                       '0',
    "visual_odometry":           'false',
    "icp_odometry":              'false',
    "subscribe_scan":            'true',
    "sync_queue_size":           '50',
    "topic_queue_size":          '50',
    "localization":              'true', # Use
localization mode, provide map matching
    "--Mem/IncrementalMemory":  'false',
    "log_level":                 'info'
}.items(),
)
]
)

```